Tire Wear and Brake Wear Trade-offs in Electric Ride-Hailing versus Personal-Use Vehicles in Urban Vancouver

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Abstract

Electric vehicles are becoming more prevalent, and this can be seen in ride-hailing fleets such as Uber and Lyft. This is seen as positive, due to eliminating exhaust emissions and reducing brake wear. However, there is increased tire wear due to heavier weight, and the stop-and-go driving style of ride-hailing fleets differs from that of a normal consumer. This study proposes comparing tire and brake wear rates between 20 ride-hailing EVs and 20 personal-use EV in urban Vancouver. All vehicles will use the same tires and brakes, which will be weighed monthly in order to quantify wear. Regression analysis will be conducted on acceleration and tires, and brakes and deceleration, along with t-tests to compare mean wear rates between groups. Findings can provide clearer information for Vancouver policy regarding fleet electrification, and provide insight into what questions should be researched next.

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1 Introduction

1.1 Background

Urban air quality has been a concern in transportation policy and in society as a whole for quite some time now, and motor vehicles are known to be a significant source of particulate matter (PM) in these urban environments. Thus, much of the historical regulations has been focusing on C02 emissions. While of course, objectively reducing C02 is helpful all around for air quality, and the environment as a whole, there are non-exhaust sources of PM that have seemingly undeservedly received less attention. In fact, tire wear, brake wear, and road surface abrasion are now beginning to be recognized as contributors to total vehicular PM. A review was carried out on tire wear particles and it was found that they made up a significant but somewhat unrecognized source of microplastics in the environment, with per capita emission rates ranging from 0.23 to 4.7 kg/year depending on the country (Kole et al., 2017). For brake matter, Grigoratos and Martini (2015) reported that brake wear can contribute up to 21% of traffic-related PM10 in urban environments, with contributions decreasing in freeways where braking is less frequent, which shows stop-and-go driving is a factor in elevated brake wear emissions.

The transition to electric vehicles (EVs) has been promoted as being key for reducing vehicle emissions as a whole, as EVs have no exhaust emissions, and also reduce brake wear by using regenerative braking. EVs are poised to make up 40% of car sales by 2030 if today's policies are maintained (IEA, 2025). However, EVs do come with downsides. Timmers and Achten (2016) found that EVs are normally around 24% heavier compared to internal combustion engine vehicles (ICEVs) because of battery mass, and weight has a positive linear relationship with certain non-exhaust emissions.

For ride-hailing fleets in particular, this is important, as they are increasingly taking up a substantial amount of miles driven in urban cities, and even without this increase, they already are responsible for a significant portion of miles, and all of this without mentioning that their stop-and-go driving is seen as worse for non-exhaust emissions then regular driving styles (Grigoratos and Martini, 2015). Jenn (2020) analyzed Uber and Lyft data from California and found that replacing a ride-hailing ICEV with an EV led to approximately a 3x emission reduction compared to replacing a personal vehicle, mostly due to the higher mileage of ride-hailing vehicles. Zhang et al. (2025) confirmed this and demonstrated that the daily carbon emission reductions from ride-hailing electrification in Chengdu, China was like eliminating 133,000 ICEV ride-hailing trips. These carbon benefits are established, and are especially valuable in Vancouver, where the hydroelectric grid gives a low-carbon energy source, however the effects of non-exhaust PM, specifically in ride-hailing fleets, remains somewhat unexamined.

1.2 Research Gap

EV adoption is becoming much more mainstream, and both government policy and consumer attitudes reflect this. However, studies on ride-hailing fleets focus mostly on C02 and economic outcomes, overlooking non-exhaust PM trade-offs. Studies are needed to examine non-exhaust trade-offs that come up when ride-hailing fleets, with their unique driving patterns, transition to heavier electric vehicles. Acceleration will be examined, as ride-hailing fleets need to stop more in order to pick up passengers.

1.3 Research Questions

1. In the urban environment of Vancouver City, to what extent does the frequent acceleration in ride-hailing drive cycles correlate with increased tire wear in electric vehicles compared to personal-use electric vehicles?
2. In the urban environment of Vancouver City, to what extent does the frequent deceleration in ride-hailing drive cycles correlate with increased brake wear in electric vehicles compared to personal-use electric vehicles?

2 Methods

2.1 Data Source

This study will use 20 ride-hailing EV drivers from platforms such as Uber or Lyft, and 20 personal-use EV drivers, from Vancouver. To ensure that vehicle-specific variables are minimized, a single EV model common to these groups will be targeted. All vehicles in the study will be fitted with the same tire make and model, and new or similarly conditioned brake pads that are made of the same compound, similar to Woo et al. (2022). GPS logging devices should already be present in each of ride-hailing vehicles, similar devices will be installed in the regular-use vehicles. These devices will provide granular speed and position data, which can be used to determine acceleration and deceleration frequency.

2.2 Data Collection

Tire wear loss will be calculated using a scale, with an initial weighing and then with more measurements being taken once a month for 12 months, adapting the weighing approach from Woo et al. (2022).

Brake wear loss will also be calculated using a scale, with weighing at the start, adapting the weighing approach from Garg et al. (2000), and with additional measurements being taken at once a month for 12 months.

2.3 Data Analysis

All analysis will be conducted with Excel. First, the dependent variable is measured tire wear rates in milligrams per vehicle per kilometer (mg/veh/km), and it will be regressed against the independent variable, acceleration frequency in events/km to assess the strength of the correlation between frequent acceleration and tire wear. Slope will tell us how much additional tire wear comes from each additional acceleration event, and the R^2 will tell us how much of the variation in tire wear between vehicles is explained by acceleration frequency. This addresses part of research question 1 (RQ1), isolating the effect of the high-frequency acceleration characteristic of ride-hailing drive cycles in urban Vancouver.

Second, the dependent variable is measured brake wear rates in mg/veh/km, will be regressed against the independent variable deceleration frequency in events/km to assess the strength of the correlation between frequent deceleration and brake wear. Slope will tell us how much additional brake wear comes from each additional deceleration event, and the R^2 will tell us how much of the variation in brake wear between vehicles is explained by deceleration frequency. This addresses part of RQ2, isolating the effect of the high-frequency deceleration characteristic of ride-hailing drive cycles in urban Vancouver.

After this, two two-tailed t-tests will also be conducted at 95% confidence level to determine whether mean tire/brake rates differ significantly between ride-hailing and personal-use EVs in urban Vancouver.

3 Implication

The results of this study will have significant implications. It is important to realize that trade-offs between decreased brake wear and increased tire wear due to ride-hailing electrification may not be the same for every community, so it is important to do Vancouver specific studies. For example, tire wear can enter streams and have adverse effects (Tian et al., 2021). Vancouver's streams lead to salmon habitats, and these salmon can be affected negatively by some of the tire wear. For reference it was discovered that 6PPD-quinone, a tire-rubber related chemical, is a key reason for the death of coho salmon that are exposed urban stormwater runoff (Tian et al., 2021). It is also important to understand which groups this affects. In Los Angeles, it was found that Black and Hispanic communities were affected disproportionately by near-roadway PM2.5 (Wen et al., 2024). Further studies will be needed to establish what quantity of brake and tire wear becomes airborne PM in Vancouver.

4 Timeline

Figure 1. The estimated time breakdown for the study. The study is estimated to have a total length of 2.5 years.

5 Budget

Table 1. This is the estimated budget breakdown for the study. The total is $48,500, and includes cost of labor, measuring equipment, and car parts.

Item	Price (CAD)	Purpose
Measurements		
Scale	$150	Weighing brake pads and tires.
GPS loggers	20 vehicles * $150 = $3,000	Tracking commuter vehicles, ride-hailing should have their own.
Standardized parts		
Brake pads	40 vehicles * $100 = $4,000	Standardizing brake pads.
Tires	40 vehicles * $300 = $12,000	Standardizing tires.
Labor		
Grad student	$10,000	Stipend for grad student.
Mechanic student	40 vehicles * 12 measurements * $30 = $14,400	Pay mechanic for brake measurements
Personal	$5,000	Personal payment for carrying out the study.
Analysis		
Excel	$0*	Used for data analysis.
Laptop	$0*	Used for data analysis.
Total	**$48,550**	

*Items provided by supervisor

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Zhang, Z., Yu, Q., Gao, K., He, H.-D., Liu, Y., & Huang, H. (2025). Carbon emission reduction benefits of ride-hailing vehicle electrification considering energy structure. *Applied Energy, 377*, 124548. doi:10.1016/j.apenergy.2024.124548

6 AI Statement

At the end of the assignment, I told ChatGPT "I care about spelling errors, please list them out". It gave me a few spelling errors, and that was it. I find that Microsoft Word is not reliable for spellcheck.

AI didn't enhance the assignment for me and didn't enhance my thinking.

The above was for my draft, after this point I did not use AI.

7 Response to Feedback

Changed all headings to bold, black, and then made them font size 12 as my text is font size 11.

From sentence "Findings can hopefully provide clearer information…)" Remove hopefully.

In my first paragraph, I state that brake wear contributions decrease on freeways where braking is less frequent. TA asked about impact of speed. Frequency is more important, as least in that example. It is well established that for example 100,000km of city driving is worse for your brake pads than the same distance on the highway, so I am not going to change that part.

Defined ICEV

Remove the sentence "The research gap is clear"

TA asked for an "and" in the last sentence in the research gap, an "and" would not have worked, but the sentence was poorly written so I rephrased it.

TA asked for justification why acceleration was picked, added to the research gap paragraph.

Clarified that I'm talking about Vancouver City

Defined mg/veh/km

Added hyphens to two two-tailed t-tests

Removed "this is a gantt chart"

Said that tire wear can enter streams before getting into the effects on salmon.

From peers:

Changed "6. References" to "References"

I had an extra gap between paragraphs, I removed it.

Peer asked me for my affiliation, but I already have UBC and ENVR 300 on the title page.

Peer asked me to move abstract below table of contents, I think that abstract should be first though.